# Construction of the EURIBOR 6M Curve: Dual-Curve Bootstrap on €STR Discounting and Tenor Basis

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October 2026

**Abstract**

This paper builds the EURIBOR 6M projection curve, the curve that gives the expected fixings of the floating leg of most euro interest rate swaps. Unlike €STR, EURIBOR is a term rate that contains a bank credit and liquidity premium, so its forwards cannot be read from the €STR curve: a separate projection curve is needed, while the cash flows are still discounted on the €STR curve built in the companion paper. We recall the EURIBOR methodology and conventions, the pricing of FRAs and swaps in the dual-curve framework, and why the floating leg no longer telescopes. The curve is bootstrapped given the €STR discount curve [**TODO:** from proxy quotes: €STR proxy curve plus a EURIBOR–€STR basis measured from the published fixings, since EURIBOR swap quotes and daily fixings are not free].

[**TODO:** Main results: repricing and validation, EURIBOR–€STR basis term structure, single-curve vs dual-curve pricing differences, 3M/6M tenor basis, par-point risks to both curves.] This paper is linked to my GitHub, which can be found on my portfolio[1], and all scripts are available in the same repository.

[1] https://portfolio-eliasbenguedda.fr

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# 1 Introduction

## 1.1 Motivation

[**TODO:** EURIBOR is the reference index of most euro swaps, swaptions, caps and floors, floating-rate loans and structured products (callable range accruals). Before 2008 a single curve (the yield curve paper of this portfolio) both discounted and projected; since then the spreads between EURIBOR and overnight rates are material, and the market uses one discount curve (€STR for collateralized trades) and one projection curve per tenor (1M, 3M, 6M, 12M).]

[**TODO:** What changes with respect to the €STR curve: term rate fixed in advance instead of an overnight rate compounded in arrears; credit and liquidity premium; projection only (discounting stays on €STR); no step forwards on ECB dates, since a 6-month rate averages the policy path over six months; tenor-specific conventions.]

## 1.2 Organization of the paper

Section 2 presents the EURIBOR rate and the dual-curve pricing of FRAs and swaps. Section 3 describes the data. Section 4 measures the EURIBOR–€STR basis and builds the proxy quotes. Section 5 bootstraps a single curve as a baseline. Section 6 builds the EURIBOR 6M projection curve on €STR discounting and validates it. Section 7 measures the pricing impact of the dual-curve framework. Section 8 adds the 3M curve and the tenor basis. Section 9 studies the par-point risk to both curves. Section 10 discusses the limitations, Section 11 lists next steps and Section 12 concludes. Appendix A gives excerpts of the Python implementation and Appendix B the notation.

# 2 The EURIBOR Rate and the Dual-Curve Framework

## 2.1 The EURIBOR fixing

[**TODO:** Administered by EMMI; hybrid methodology since 2019 (level 1: transactions, level 2: interpolations, level 3: model-based); panel of banks; tenors 1W, 1M, 3M, 6M, 12M; published at 11:00 CET for value spot $(T+2)$. The rate of an unsecured term deposit between banks: credit and liquidity premium over the expected overnight rate.]

## 2.2 Term rate vs overnight rate

A EURIBOR fixing $L(T_s, T_e)$ is fixed at the start of the period and paid at its end. The €STR leg pays the overnight rate compounded in arrears, whose value is $P_d(0,T_s) - P_d(0,T_e)$ (companion paper). The difference

$$L(T_s,T_e) - R^{\text{€STR}}(T_s,T_e) \tag{1}$$

is the term basis: the premium for lending for six months to a bank rather than overnight. [**TODO:** Order of magnitude, behaviour in stress periods (2008, 2011–2012, 2022).]

## 2.3 Projection curve and forwards

The projection curve is described by pseudo-discount factors $P_6(0,t)$, which are not prices but a convenient parametrization of the forward EURIBOR 6M rates:

$$L_j(0) = \frac{1}{\delta_j}\Big(\frac{P_6(0,t_{j-1})}{P_6(0,t_j)} - 1\Big), \qquad \delta_j = \frac{t_j - t_{j-1}}{360}. \tag{2}$$

[**TODO:** $L_j(0)$ is the expectation of the fixing under the $t_j$-forward measure associated with €STR discounting (no convexity adjustment for a payment at the end of the period).]

## 2.4 FRA and swap pricing

A FRA on $[T_1,T_2]$ has fair rate $F = L(0;T_1,T_2)$ from (2). A payer swap paying a fixed rate $K$ on the schedule $\{t_k^{\text{fix}}\}$ and receiving EURIBOR 6M on $\{t_j\}$ has par rate

$$S_6(T) = \frac{\sum_j \delta_j\, L_j(0)\, P_d(0,t_j)}{\sum_k \delta_k^{\text{fix}}\, P_d(0,t_k^{\text{fix}})}. \tag{3}$$

In the single-curve case $P_6 = P_d$, the floating leg telescopes to $P_d(0,t_0) - P_d(0,t_n)$; in the dual-curve case it does not, and the projection curve has to be bootstrapped given the discount curve.

## 2.5 Conventions

| Item | EURIBOR 6M convention |
|---|---|
| Calendar | TARGET2 |
| Fixing | 11:00 CET, two business days before the start |
| Spot lag | $T + 2$ business days |
| Floating leg | EURIBOR 6M, semi-annual, ACT/360 |
| Fixed leg | annual, 30/360 [**TODO:** check] |
| Business day convention | modified following, end-of-month rule [**TODO:** check] |
| Short swaps | 1Y–2Y swaps often quoted against EURIBOR 3M [**TODO:** check] |
| Discounting | €STR (cleared swaps, cash collateral in EUR) |

Table 1: Market conventions of EURIBOR 6M swaps. [**TODO:** Verify against the ISDA / market documentation.]

# 3 Data

## 3.1 EURIBOR fixings

[**TODO:** `Euribor_Monthly.csv`: monthly averages of the daily fixings for 1M, 3M, 6M, 12M since January 2015 (ECB, `FM` dataset). Daily fixings are published by EMMI under licence and are not free: the fixing on the valuation date is not available exactly.]

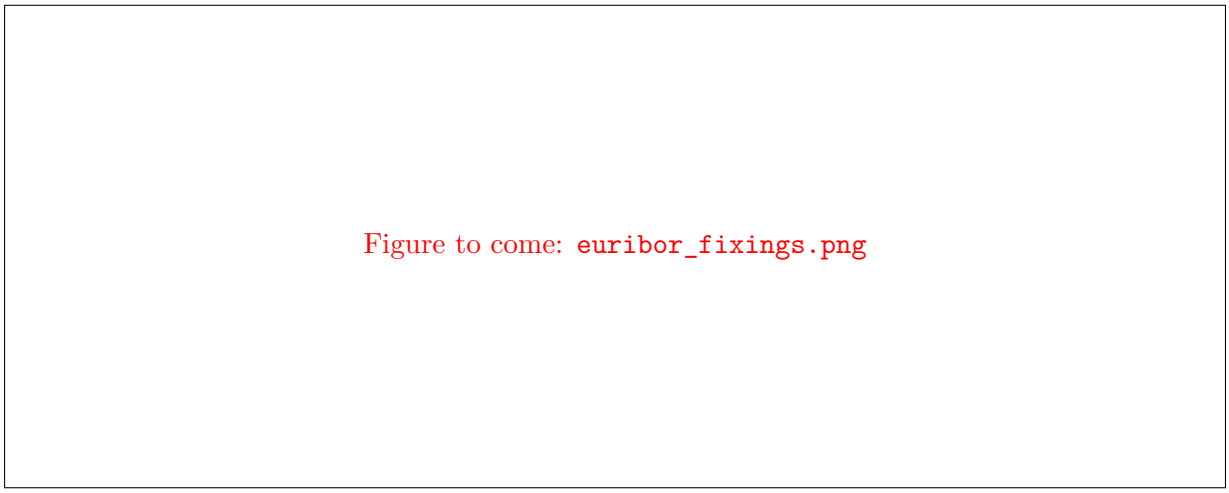



Figure 1: EURIBOR 1M, 3M, 6M, 12M (monthly averages) and €STR. [**TODO:** ]

## 3.2 Discount curve: €STR

[**TODO:** €STR curve of the companion paper (meeting steps up to 2 years, raw beyond), same valuation date; built on proxy quotes.]

## 3.3 Market quotes

[**TODO:** EURIBOR 6M swaps, FRAs and EURIBOR 3M futures are not free. Template `EURIBOR6M_Quotes_TEMPLATE.csv` to fill by hand; otherwise proxy quotes (Section 4). The EURIBOR 6M zero curve of 12/02/2025 used in the Heston and yield curve papers is a single-curve curve at another date.]

| Instrument | Rate (%) | Instrument | Rate (%) | Instrument | Rate (%) |
|---|---|---|---|---|---|
| 6M fixing | [**TODO:** ] | FRA $12 \times 18$ | [**TODO:** ] | IRS 5Y | [**TODO:** ] |
| FRA $1 \times 7$ | [**TODO:** ] | IRS 2Y | [**TODO:** ] | IRS 10Y | [**TODO:** ] |
| FRA $3 \times 9$ | [**TODO:** ] | IRS 3Y | [**TODO:** ] | IRS 20Y | [**TODO:** ] |
| FRA $6 \times 12$ | [**TODO:** ] | IRS 4Y | [**TODO:** ] | IRS 30Y | [**TODO:** ] |

Table 2: EURIBOR 6M curve instruments on [**TODO:** valuation date].

# 4 EURIBOR–€STR Basis and Proxy Quotes

## 4.1 Measuring the basis

[**TODO:** Two measures, as for the government–OIS basis of the companion paper: (a) ex post, EURIBOR 6M fixing minus €STR compounded over the following six months (mixes the basis with forecast errors); (b) contemporaneous, EURIBOR 6M fixing minus the 6M €STR OIS rate of the same date (depends on the €STR proxy). Monthly averages only.]

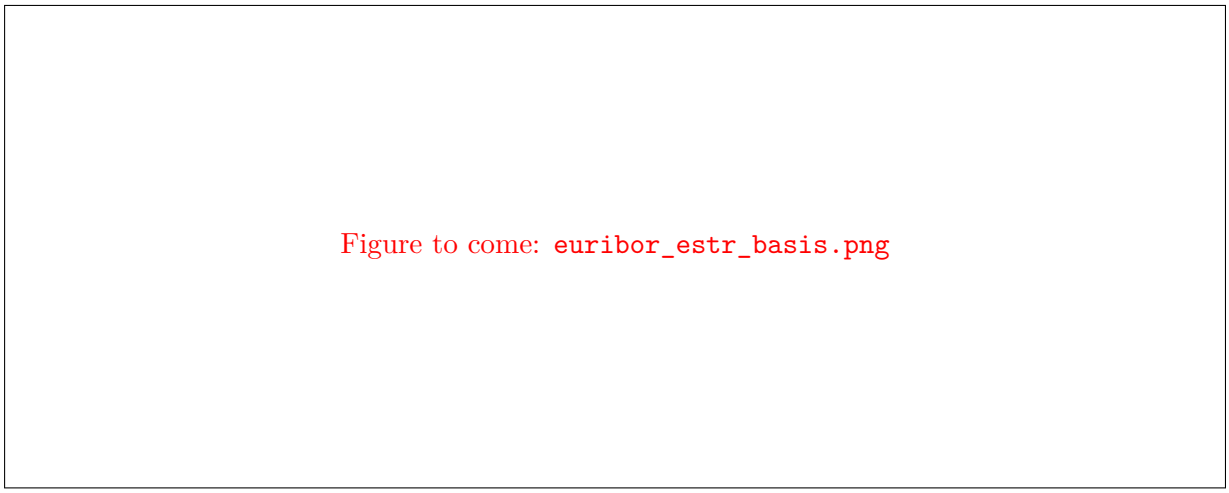



Figure 2: EURIBOR 6M minus €STR over the same period. [**TODO:** ]

| Year | 1M (bp) | 3M (bp) | 6M (bp) | 12M (bp) |
|---|---|---|---|---|
| [**TODO:** ] | | | | |

Table 3: Average EURIBOR–€STR basis by tenor and year.

## 4.2 Term structure of the basis

[**TODO:** The basis is quoted only at the fixing tenor; for longer maturities (forward 6M periods), assumption needed: flat, mean-reverting to a long-term level, or read from €STR/EURIBOR basis swaps if available.]

## 4.3 Proxy projection curve and quotes

[**TODO:** Forward EURIBOR 6M = forward compounded €STR over the period (from the €STR curve) + basis $s(t)$; proxy FRA and swap quotes = par rates (3) on this curve with €STR discounting.]

## 5 Single-Curve Bootstrap (Baseline)

[**TODO:** Same instruments bootstrapped in the single-curve framework of the yield curve paper (the EURIBOR curve discounts and projects). Baseline for Section 7; checks the instruments (FRA, IRS with 30/360 fixed and ACT/360 floating legs) with the existing `Bootstrapper`.]

| Method | Max residual (bp) | Solver | Time (ms) |
|---|---|---|---|
| Raw | **[TODO: ]** | sequential | **[TODO: ]** |
| Monotone convex | **[TODO: ]** | simultaneous | **[TODO: ]** |

Table 4: Single-curve bootstrap of the EURIBOR 6M instruments.

# 6 Dual-Curve Bootstrap on €STR Discounting

## 6.1 Instruments

| Instrument | Quote | Pricing equation |
|---|---|---|
| EURIBOR 6M fixing | rate $L_0$ | $1 + L_0\,\delta = P_6(0,T_0)/P_6(0,T_0+6\text{M})$ |
| FRA $[T_1,T_2]$ | rate $F$ | $1 + F\,\delta = P_6(0,T_1)/P_6(0,T_2)$ |
| EURIBOR 3M future | price $100-F$ | FRA rate + convexity adjustment **[TODO: ]** |
| IRS vs 6M | par rate $S_6$ | equation (3), $P_d$ fixed |

Table 5: Instruments of the EURIBOR 6M projection curve.

[**TODO:** New instrument classes holding a reference to the discount curve; `implied_quote(curve)` takes the projection curve only, so the existing `Bootstrapper` solves for $P_6$ unchanged.]

## 6.2 Pillars and interpolation

[**TODO:** Pillars at the end of the last projected period of each instrument (6M fixing, FRA end dates, swap maturities). Interpolate the projection curve itself, or the spread over the €STR forwards (the €STR steps of the ECB meetings then pass into the EURIBOR forwards, and only the smoother basis is interpolated). Comparison raw / monotone convex / spread interpolation.]

Figure to come: `projection_forwards.png`

Figure 3: Forward EURIBOR 6M rates $L(t,t+6\text{M})$ and forward compounded €STR over the same periods. **[TODO: ]**

## 6.3 Solver

[**TODO:** Sequential (raw) or simultaneous (non-local) solve for the pseudo zero rates of $P_6$, $P_d$ fixed; starting point: single-curve solution.]

## 6.4 Validation

(1) **Repricing**: all instruments, maximum residual below $10^{-6}$ bp. **[TODO: ]**

(2) **Independent repricing**: swap PVs recomputed leg by leg; comparison with QuantLib [8] (`SwapRateHelper` with an exogenous discounting curve). **[TODO: ]**

(3) **Exact round trip**: quotes generated from known $(P_d, P_6)$, projection curve recovered to machine precision. **[TODO: ]**

(4) **Consistency**: setting $P_d = P_6$ gives back the single-curve bootstrap of Section 5. **[TODO: ]**

(5) **Basis**: forward EURIBOR minus forward €STR positive and plausible at all maturities. **[TODO: ]**

| Test | Criterion | Result |
|---|---|---|
| Repricing | $\max \lvert R_k \rvert < 10^{-6}$ bp | **[TODO: ]** |
| Independent repricing / QuantLib | $<$ **[TODO:**0.01**]** bp | **[TODO: ]** |
| Exact round trip | machine precision | **[TODO: ]** |
| $P_d = P_6$ | single-curve result | **[TODO: ]** |
| Basis | positive, plausible | **[TODO: ]** |

Table 6: Validation tests of the dual-curve bootstrap.

## 7 Single-Curve vs Dual-Curve Pricing

[**TODO:** Both frameworks reprice the par swaps; they differ on everything else: off-market swaps, forward-starting swaps, FRAs between pillars, swaptions and caps (annuity and forwards). Price differences in bp of notional for a few instruments; differences between the projection curve and the single curve.]

| Instrument | Single curve | Dual curve | Difference |
|---|---|---|---|
| [**TODO:** ] | | | |

Table 7: Prices of test instruments in the single-curve and dual-curve frameworks.

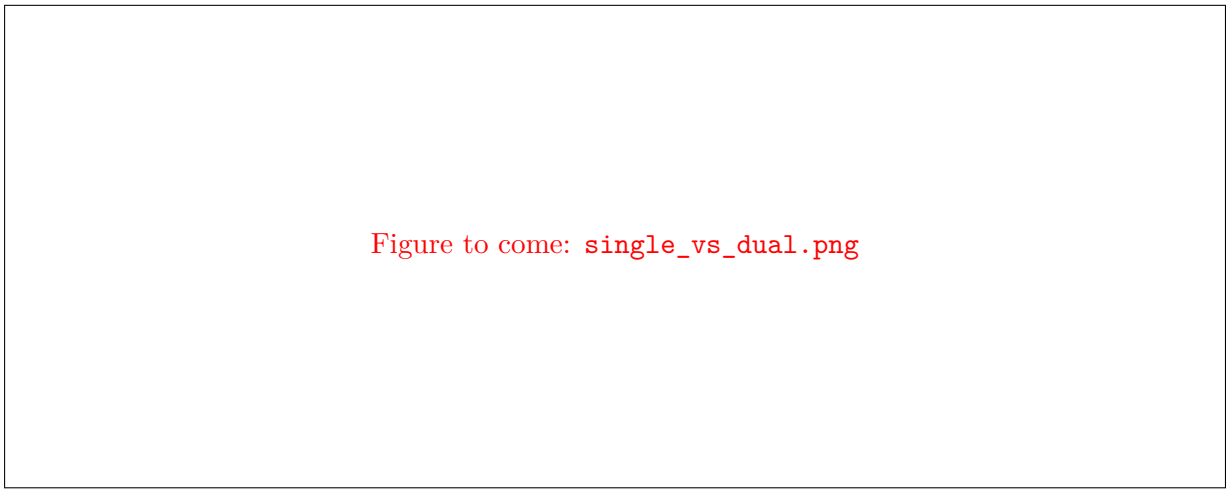



Figure 4: Single-curve EURIBOR curve vs dual-curve projection and discount curves. [**TODO:** ]

## 8 Tenor Basis: the EURIBOR 3M Curve

[**TODO:** EURIBOR 3M projection curve from 3M fixings, futures and 3s6s basis swaps; 3M/6M basis $L_6 - L_3$ (longer tenor, higher premium); short swaps quoted against 3M. Optional if the 3M quotes are not available.]

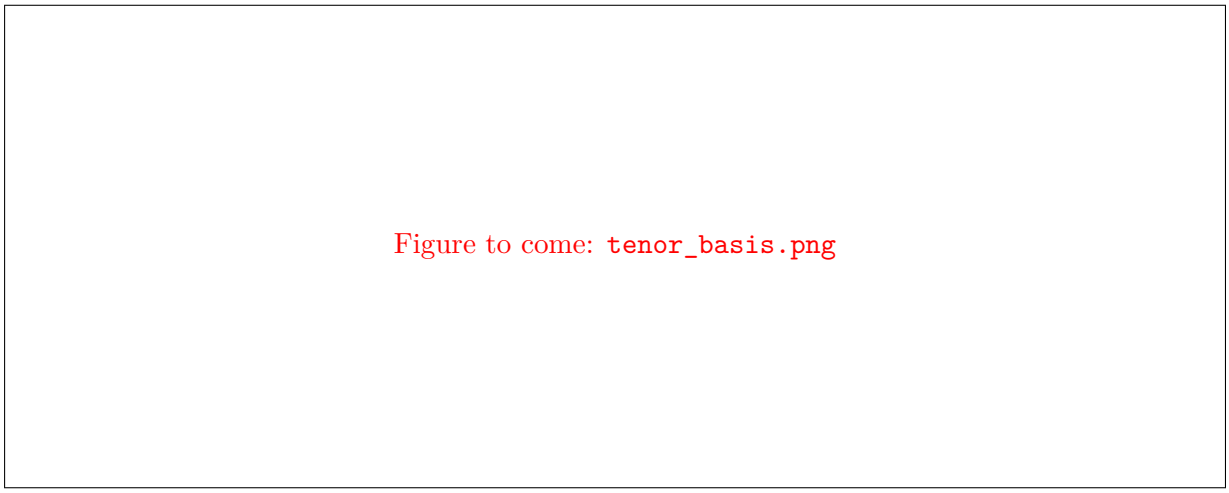



Figure 5: EURIBOR 3M and 6M forward curves and tenor basis. [**TODO:** ]

## 9 Risk: Par-Point Sensitivities to Both Curves

[**TODO:** Sensitivities of a EURIBOR 6M swap (and an 18m × 4y forward swap) to the €STR OIS quotes and to the EURIBOR quotes, by re-bootstrapping both curves; the projection curve depends on the discount curve, so a bump of an OIS quote moves both. Basis risk vs outright rate risk; Jacobian method as next step.]

| Quote bumped | DV01 (EUR) | of which via projection |
|---|---|---|
| €STR OIS [**TODO:** ]<br>EURIBOR 6M [**TODO:** ] | | |
| Parallel +1 bp (both curves) | [**TODO:** ] | |

Table 8: Par-point DV01 of a test swap to the €STR and EURIBOR 6M quotes.

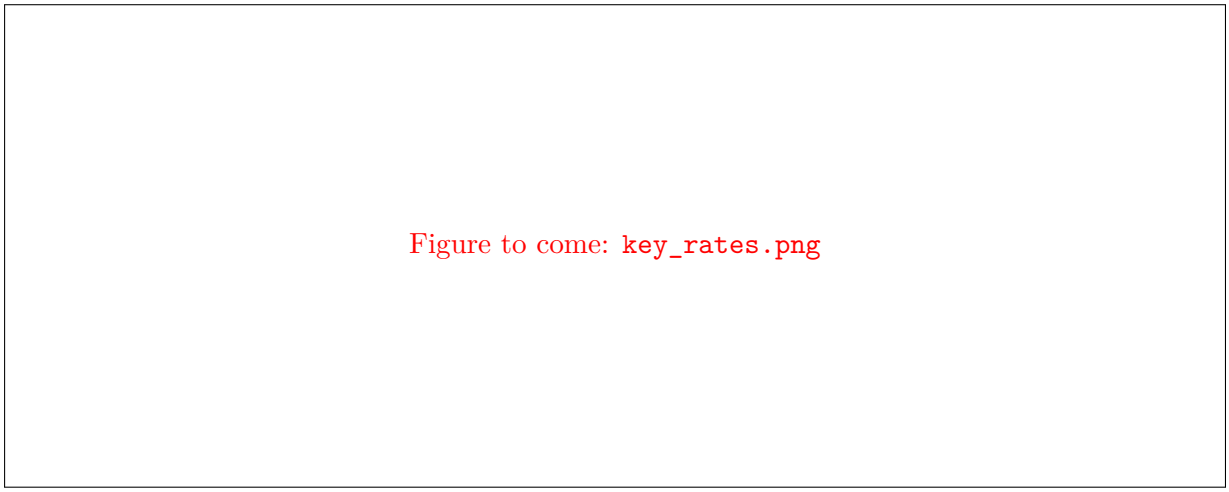



Figure 6: Par-point sensitivities to the €STR and EURIBOR 6M quotes. [**TODO:** ]

## 10 Limitations

**Proxy quotes.** [**TODO:** Proxy €STR curve plus a basis measured on monthly averages; term structure of the basis assumed.]

**Fixings.** [**TODO:** Daily EURIBOR fixings not available; today's 6M fixing approximated.]

**Futures.** [**TODO:** Convexity adjustment model-dependent (Hull–White).]

**Conventions.** [**TODO:** 30/360 fixed leg, end-of-month rule, stub periods.]

**Framework.** [**TODO:** Deterministic basis; collateral in EUR cash; EURIBOR fallback to €STR not modelled.]

## 11 Next Steps

- [**TODO:** Real quotes: fill `EURIBOR6M_Quotes_YYYYMMDD.csv`.]
- [**TODO:** EURIBOR 1M and 12M curves; basis swaps.]
- [**TODO:** Simultaneous bootstrap of the €STR and EURIBOR curves from IRS and €STR/EURIBOR basis swaps (long end of the €STR curve from EURIBOR swaps).]
- [**TODO:** Hull–White calibration on the dual-curve setup and pricing of swaptions, caps and the callable range accrual.]
- [**TODO:** Daily history and principal component analysis of the EURIBOR–€STR basis.]

## 12 Conclusion

[**TODO:** Summary of the construction, of the basis, of the single vs dual-curve differences and of the risk.]

# A Python Implementation (Excerpts)

The following listings are excerpts of the implementation (package `Fixed_Income_Models`), slightly simplified. The full scripts and the notebook `euribor_curve.ipynb` are available on GitHub.

## A.1 Dual-curve instruments

```
# TODO : FRA and IRS classes holding the ESTR discount curve, implied_quote(
    projection_curve)
```

## A.2 Basis measurement and proxy quotes

```
# TODO
```

## A.3 Dual-curve bootstrap

```
# TODO : Bootstrapper on the projection curve, discount curve fixed
```

# B Notation

| Symbol | Meaning |
|---|---|
| $P_d(0,t)$ | €STR discount factor |
| $P_6(0,t),\ P_3(0,t)$ | Pseudo-discount factors of the EURIBOR 6M and 3M projection curves |
| $L(T_s,T_e),\ L_j(0)$ | EURIBOR fixing of the period $[T_s,T_e]$ and forward EURIBOR of period $j$ |
| $R^{\text{€STR}}(T_s,T_e)$ | €STR compounded over $[T_s,T_e]$ |
| $s(t)$ | EURIBOR–€STR basis |
| $\delta_j,\ \delta_k^{\text{fix}}$ | ACT/360 floating and 30/360 fixed accrual fractions |
| $S_6(T)$ | Par rate of a EURIBOR 6M swap |
| $q_k,\ R_k,\ z$ | Market quotes, residuals and bootstrap unknowns |

Table 9: Main notation.

## References


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